UNIT 2 & 3 · TOPICS 2.12, 3.1–3.2, 4.1 · SAMPLING DISTRIBUTIONS

Sampling Distributions & Central Limit Theorem

Exact notation, parameter vs. statistic distinctions, standard error formulas, and the conditions required for Normal approximations.

1. Variable Definitions: Parameter vs. Statistic

Never mix up **parameters** (fixed, usually unknown numbers describing the entire population) and **statistics** (random variables computed from a sample that vary from sample to sample).

SYMBOL	NAME	DEFINITION & ROLE	TYPE
μ	Population Mean	True arithmetic average of all N individuals in the population.	Parameter (fixed)
$\bar{x}$	Sample Mean	Arithmetic average of a sample of size n: $\bar{x} = (1/n)\Sigma x_i$.	Statistic (random)
σ, σ^2	Population SD / Variance	True spread of individual values in the population around μ.	Parameter (fixed)
s, s^2	Sample SD / Variance	Sample spread around $\bar{x}$ using $n - 1$ in denominator (Bessel's correction).	Statistic (random)
p	Population Proportion	True fraction of successes in the population ($0 \le p \le 1$).	Parameter (fixed)
$\hat{p}$	Sample Proportion	Fraction of successes in a sample: $\hat{p} = X / n$ (X = count of successes).	Statistic (random)
n, N	Sample / Population Size	n = number of observations in sample; N = total size of population.	Counts ($n \ge 1$)
$\mu_{\bar{x}}$, $\sigma_{\bar{x}}$	Mean & SD of $\bar{x}$	Center and spread of the sampling distribution of all possible sample means of size n.	Sampling Dist.
SE	Standard Error	Estimated standard deviation of a statistic computed by plugging in s or $\hat{p}$ when σ or p is unknown.	Estimated Spread

2. Sampling Distribution of a Sample Mean ($\bar{x}$)

Suppose we take all possible Simple Random Samples (SRS) of size n from a population with mean μ and standard deviation σ.

CENTER, SPREAD & STANDARDIZATION OF $\bar{x}$

$$\mu_{\bar{x}} = \mu \quad | \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad | \quad \text{SE}_{\bar{x}} = \frac{s}{\sqrt{n}} \quad | \quad z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Variables: $\bar{x}$ = observed sample mean; μ = population mean; σ = population SD; s = sample SD; n = sample size.
Key Insight: To cut the standard deviation of $\bar{x}$ in half, you must multiply sample size n by **4**.

CENTRAL LIMIT THEOREM (CLT)

Even if the parent population distribution is skewed or non-Normal, as sample size n increases, the sampling distribution of $\bar{x}$ becomes **approximately Normal**: $\bar{x} \sim N(\mu, \sigma / \sqrt{n})$. Standard threshold used in intro/AP stats: $\boldsymbol{n \geq 30}$ (if the population is already Normal, $\bar{x}$ is *exact* Normal for any $n \geq 1$).

3. Sampling Distribution of a Sample Proportion ($\hat{p}$)

When counting binary successes ($X \sim \text{Binomial}(n, p)$) and converting to a sample proportion $\hat{p} = X / n$:

CENTER, SPREAD & STANDARDIZATION OF $\hat{P}$

$$\mu_{\hat{p}} = p \quad | \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad | \quad \text{SE}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad | \quad z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

Variables: p = true population proportion; $\hat{p} = X/n$ = observed sample proportion; $1 - p$ (often denoted q) = probability of failure; n = sample size.

4. Two-Sample Sampling Distributions (Differences)

For two **independent** random samples of sizes n_1 and n_2, the mean is the difference of centers, while **variances ALWAYS add** (never subtract variances!).

STATISTIC	CENTER (MEAN)	TRUE STANDARD DEVIATION (Σ)	STANDARD ERROR (SE)
Difference of Means $\bar{x}_1 - \bar{x}_2$	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Difference of Proportions $\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

5. Required Conditions Checklist (R-I-N)

CONDITION	WHAT IT GUARANTEES	HOW TO CHECK (MEANS VS. PROPORTIONS)
1. Random	Unbiasedness: $\mu_{\bar{x}} = \mu$ or $\mu_{\hat{p}} = p$	Data collected from an SRS, stratified/cluster random sample, or randomized experiment.
2. Independent (10% Rule)	Validates spread formulas ($\sigma/\sqrt{n}$) when sampling without replacement	Check $n \leq 0.10N$ (or $N \geq 10n$). Note: Not needed for randomized experiments with random assignment!
3. Normal / Large Sample	Normal shape so z-table / `normalcdf` applies	**Means:** Population is Normal, OR $n \geq 30$ (CLT), OR sample plot has no strong skew/outliers. **Proportions (Large Counts):** $np \geq 10$ and $n(1-p) \geq 10$ (use $\hat{p}$ or p_0 depending on context).

6. Common Traps & Calculator Syntax

COMMON MISTAKES

- Using σ instead of $\sigma/\sqrt{n}$ when asked about a *sample mean* $\bar{x}$ (read the prompt: one individual vs. average of n individuals!).
- Subtracting variances: $\sigma_{\bar{x}_1-\bar{x}_2} \neq \sqrt{\sigma_1^2/n_1 - \sigma_2^2/n_2}$. Variances **always add**.
- Applying the CLT ($n \geq 30$) to *proportions*. Proportions require the **Large Counts** check ($np \geq 10$, $n(1-p) \geq 10$).

CALCULATOR & EXAM QUICK CHECK

- **Probability for $\bar{x}$:** `normalcdf(lower, upper, μ, σ/√n)`
- **Probability for $\hat{p}$:** `normalcdf(lower, upper, p, √(p(1−p)/n))`
- **Unbiasedness:** A statistic is unbiased if its sampling distribution mean equals the parameter ($E(\bar{x}) = \mu$).
- **Precision vs. Accuracy:** Larger n lowers spread (higher precision), while random sampling prevents bias (accuracy).

7. Estimators, Bias and Variability (Topic 3.1)

A sample statistic used as a single best guess of a population parameter is a **point estimator**. The number it produces from one sample is the **estimate**. For example $\hat{p}$ estimates the population proportion p, and $\bar{x}$ estimates the population mean μ.

UNBIASEDNESS

$$\text{estimator is unbiased} \Leftrightarrow \text{E(estimator)} = \text{parameter}$$

An estimator is unbiased if, **on average over all possible random samples**, it neither systematically underestimates nor overestimates the parameter. Equivalently, the mean of its sampling distribution equals the parameter: $\text{E}(\hat{p}) = p$ and $\text{E}(\bar{x}) = \mu$. Unbiasedness is a property of the *procedure*, not of any single estimate.

PROPERTY	WHAT IT MEASURES	CONTROLLED BY
Bias (accuracy)	Systematic tendency to miss the parameter in one direction; the centre of the sampling distribution	**Sampling design** — use random sampling / random assignment; a larger sample does not fix bias
Variability (precision)	Spread of estimates across repeated samples, measured by the **standard error**	**Sample size** — larger n shrinks SE, e.g. $\sigma/\sqrt{n}$

TRAPS THE EXAM ACTUALLY TESTS

- **Unbiased does not mean "correct every time"** — individual estimates still vary; it means there is no systematic error in the long run.
- **A larger sample reduces variability, not bias.** A biased sampling method (convenience sample, voluntary response) stays biased no matter how large n is.
- **Keep the two goals separate**: random selection addresses accuracy (bias); sample size addresses precision (variability).
- **Name the estimator and the parameter in context** when justifying unbiasedness, and refer to the *sampling distribution*, not one sample's result.